

BrokerVerse OOTB

Insurance broking platform by iorta TechNXT

Solution Architecture · Document 02 of 11

Database Design

Product	BrokerVerse OOTB
Document	02 Database Design
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Document control

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Version history

Version	Date	Author	Change
1.0	29 September 2026	iorta TechNXT	Initial issue
1.1	29 September 2026	iorta TechNXT	Updated to the current system: placement journey, co-insurance, posting rules, period end and BIR tax, bank reconciliation, document numbering, commission rate matrix, single PDF engine, uploads and go-live imports, housekeeping, business time zone, broker roles, deployment on the BrokerVerse URL

Review and approval

Role	Name	Signature	Date
Solution architect			
DevOps / infrastructure lead			
Information security			
Business owner			

Conventions

- Statements of fact are derived from the code, configuration and database of the baseline above (file paths are given relative to the repository). Figures measured on the local test system are labelled **test data** or **indicative**.
- Targets, sizes, retention periods and thresholds that are not facts of the code are labelled **Recommended / to be confirmed by the business and DevOps**.
- Items marked **Gap** are defects or missing capabilities found while preparing this document.

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1 Introduction

1.1 Purpose and scope

This document describes the logical and physical design of the BrokerVerse OOTB database: the schema by business domain, the entity-relationship structure of each domain, the keys, the JSONB document columns, document numbering, audit and soft-delete conventions, the posting rules and period controls that protect the ledger, the migration and seed approach, and the handling of time zones, dates and amounts. The table-by-table inventory with sizes, row counts and retention classes is document 03.

Sources: `backend/src/db/migrations/*.sql` (58 files, 3,770 lines), `backend/src/db/seeds`, `backend/src/db/pool.js`, `migrate.js`, `seed.js`, `backend/src/lib/dates.js`, `money.js`, `numbering.js`, and the catalogue of a PostgreSQL 16 database created for this document (`brokerverse_arch`: migrated and seeded with reference and sample data; snapshot in `docs/architecture/tools/data/db_snapshot.json`).

1.2 Database platform

Item	Value (local test database)
PostgreSQL version	16.13 (Ubuntu 16.13-0ubuntu0.24.04.1)
Database size	20.7 MB (test data)
Server time zone	Asia/Manila
Tables (schema public)	138
Columns by type	text 1320, timestamp with time zone 310, numeric 256, integer 139, date 99, jsonb 92, bigint 47, boolean 44, ARRAY 5
Indexes	448 (229 unique incl. 138 primary keys, 33 partial, 1 GIN)
Foreign keys	234 (36 without a supporting index)
Identity sequences (serial / identity columns)	63
Document-number counters (table sequences)	24 counters for 2026
Extensions	pgcrypto 1.3, plpgsql 1.0
Application functions	<code>accounting_period_lock_guard</code> , <code>backfill_risk_participants</code> , <code>document_numbering_mirror_prefix</code> , <code>format_document_number</code> , <code>gl_accounts_defaults</code> , <code>jv_check_posting</code> , <code>next_document_number</code> , <code>next_number</code> , <code>numbering_business_date</code> , <code>numbering_fiscal_year</code> , <code>numbering_optional_token</code> , <code>numbering_period_key</code> , <code>numbering_prefix</code> , <code>pe_balance_before</code> , <code>quotes_link_product</code> , <code>rp_insurer_id</code> , <code>rp_parts_from_doc</code> , <code>rp_share</code> , <code>rp_write_split</code> , <code>rpt_age_bucket</code> , <code>set_updated_at</code> , <code>user_effective_roles</code>
Triggers	<code>accounting_period_lock_guard_trg</code> (<code>accounting_periods</code>), <code>commission_rates_updated</code> (<code>commission_rates</code>), <code>document_numbering_mirror</code> (<code>document_numbering</code>), <code>document_numbering_updated</code> (<code>document_numbering</code>), <code>gl_accounts_defaults_trg</code> (<code>gl_accounts</code>), <code>jv_check_posting_trg</code> (<code>journal_vouchers</code>), <code>quotes_link_product</code> (<code>quotes</code>), <code>roles_updated</code> (<code>roles</code>), <code>users_updated</code> (<code>users</code>)
Views / materialised views	<code>bank_account_links</code> , <code>bank_book_lines</code>
Migrations applied	58 (0001_core.sql to 0152_email_sender.sql)
Application settings (<code>app_settings</code>)	351 keys in 36 groups
Scheduled jobs (<code>scheduled_jobs</code>)	15
Snapshot taken	2026-09-29T16:04:56+00:00

One database and one schema (`public`) hold all data. The application connects with a single login through `DATABASE_URL` (node-postgres pool, maximum 10 connections per API process). The login needs the right to create tables because the API applies its own migrations on start. The database time zone is set to `Asia/Manila` (`ALTER DATABASE ... SET timezone`, or the RDS parameter group), as `deploy/README.md` requires. Only the `pgcrypto` extension is used, for `gen_random_bytes()` in primary-key defaults.

2 Schema overview by domain

The 138 tables fall into 18 domains. Row counts and sizes are from the database created for this document (reference data plus the sample data set) and are **test data**.

Domain	Tables	Rows (test data)	Size KB (test data)	Tables
Security and identity	9	240	312	login_history, password_history, password_resets, permissions, refresh_tokens, role_permissions, roles, user_roles, users
Platform services	14	872	1,120	agent_events, app_settings, audit_log, document_numbering, documents, email_outbox, job_queue, job_runs, master_records, master_types, notifications, scheduled_jobs, schema_migrations, sequences
Reference masters	16	234	704	banks, branches, cities, countries, coverages, currencies, districts, insurance_companies, policy_types, postal_codes, products, signatories, states, vehicle_brands, vehicle_models, vehicle_variants
Product configurator	4	73	280	product_components, product_risk_mappings, product_risk_sections, product_templates
Sales and policy	6	81	872	clients, endorsements, leads, policies, policy_payments, quotes
Placement and co-insurance	4	52	528	broker_slips, insurer_offers, placements, risk_participants
Claims	4	111	248	claim_field_changes, claim_history, claim_settlement_movements, claims
Renewals	7	54	448	renewal_activities, renewal_batch_policies, renewal_batches, renewal_notices, renewal_quotes, renewals, winback_campaigns
Billing, receipts and collections	9	134	696	collection_actions, collection_items, receipt_applications, receipt_lines, receipts, receivable_credits, receivable_participants, receivables, write_off_reasons
General ledger and posting rules	7	556	744	accounting_periods, entry_matches, gl_accounts, journal_lines, journal_vouchers, posting_rule_lines, posting_rules
Period end and tax	12	37	392	bir_2307_certificates, fiscal_years, opening_balances, period_close_checklist, period_close_entries, period_close_run_checks, period_close_runs, period_status_history, recurring_journal_runs, recurring_journals, tax_codes, year_end_runs
Bank reconciliation	9	27	384	bank_match_rules, bank_rec_match_items, bank_rec_matches, bank_reconciliation_history, bank_reconciliations, bank_statement_formats, bank_statement_lines, bank_statements, bank_transaction_types
Disbursements and petty cash	9	32	768	checkbooks, disbursements, invoice_lists, petty_cash_disbursements, petty_cash_funds, petty_cash_receipts, petty_cash_replenishments, petty_cash_request_lines, petty_cash_requests
Commission	4	30	288	commission_adjustments, commission_rates, commission_referrers, commissions
Remittance and direct bill	11	89	696	commission_debit_note_collections, commission_debit_note_lines, commission_debit_notes, direct_bill_items, insurer_refund_credits, remittance_allocations, remittance_approvals, remittance_delegations, remittance_items, remittance_lines, remittances
Reinsurance	7	37	480	cessions, reinsurance_bordereaux, reinsurance_exceptions, reinsurance_reconciliations, reinsurance_recoveries, reinsurance_treaties, reinsurers
Incentive	3	26	160	incentive_calculations, incentive_programs, incentive_results
Reporting	3	39	192	generated_reports, report_definitions, report_schedules
Total	138	2,724	9,312	

3 Design conventions

3.1 Keys

Convention	Detail
Business records	54 tables use a text primary key made of a readable prefix and random hexadecimal characters generated by the database, for example <code>policies.id</code> <code>DEFAULT 'pol_' encode(gen_random_bytes(8), 'hex')</code> . Prefixes include <code>ld_</code> lead, <code>cl_</code> client, <code>qt_</code> quotation, <code>bs_</code> broker slip, <code>ofr_</code> insurer offer, <code>plc_</code> placement, <code>pol_</code> policy, <code>end_</code> endorsement, <code>clm_</code> claim, <code>rnw_</code> renewal, <code>or_</code> receipt, <code>jv_</code> journal voucher, <code>pv_</code> payment voucher, <code>rm_</code> remittance, <code>dn_</code> debit note, <code>brc_</code> bank reconciliation, <code>yec_</code> year-end run, <code>usr_</code> user and <code>doc_</code> document. The ids cannot be guessed and do not collide across tables.
Reference, line and log tables	63 tables use integer or bigint serial keys: masters (countries, banks, insurers, products), lines and logs that grow fast (<code>journal_lines</code> , <code>audit_log</code> , <code>login_history</code> , <code>job_runs</code> , <code>claim_history</code> , <code>risk_participants</code> , <code>bank_rec_match_items</code> , <code>period_status_history</code>).
Natural and composite keys	21 tables: <code>app_settings(key)</code> , <code>gl_accounts(code)</code> , <code>master_types(code)</code> , <code>report_definitions(code)</code> , <code>accounting_periods(period)</code> , <code>fiscal_years(code)</code> , <code>tax_codes(code)</code> , <code>bank_statement_formats(code)</code> , <code>bank_transaction_types(code)</code> , <code>bank_match_rules(code)</code> , <code>period_close_checklist(code)</code> , <code>opening_balances(fiscal_year, account_code)</code> , <code>receivable_participants(receivable_id, insurance_company_id)</code> , <code>refresh_tokens(jti)</code> , <code>schema_migrations(name)</code> , <code>sequences(name, period)</code> , <code>role_permissions</code> , <code>user_roles</code> and others.
Business numbers	Readable document numbers (<code>QT-2026-00001</code> , <code>POL-2026-00001</code> , <code>OR-2026-00019</code> , <code>BS-</code> , <code>PS-</code> , <code>MEC-</code> , <code>BRC-</code>) are separate unique columns (<code>quote_number</code> , <code>policy_number</code> , <code>receipt_number</code> , <code>slip_number</code> , <code>run_number...</code>), never primary keys. Most detail endpoints accept either the id or the number.
Foreign keys	234 foreign keys: 200 <code>NO ACTION</code> (a referenced record cannot be deleted while in use), 30 <code>CASCADE</code> (children with no meaning without their parent: journal lines, receipt lines, match items, claim history, renewal notices, role grants, refresh tokens, notifications) and 4 <code>SET NULL</code> .

3.2 Column types

Data	Type and handling
Amounts	<code>numeric</code> (256 columns), typically <code>numeric(14,2)</code> ; converted to JavaScript numbers by the pool type parser; all rounding through <code>round2()</code> in <code>lib/money.js</code> (half away from zero, the same rule as PostgreSQL <code>round(numeric, 2)</code>). Shares are <code>numeric(7,4)</code> .
Business dates	<code>date</code> (99 columns: inception, expiry, due, loss, receipt, voucher, statement and period dates), returned to the application as <code>YYYY-MM-DD</code> strings so no time-zone shift can occur. Date columns that default to today take <code>numbering_business_date()</code> , the date in <code>general.timezone</code> .
Instantns	<code>timestamptz</code> (310 columns, for example <code>created_at</code> , <code>approved_at</code> , <code>sent_at</code>); there are no <code>timestamp without time zone</code> columns.
Flexible parts	<code>jsonb</code> (92 columns in 60 tables), see below.
Codes and statuses	<code>text</code> with validation in the application (zod, service rules) and 91 <code>CHECK</code> constraints; statuses are lower-case codes mapped to display labels.

3.3 JSONB document columns

JSONB is used where the structure varies by product, screen or configuration. Columns that are searched, joined, totalled or reported on stay relational.

Table	JSONB columns and use
quotes, broker_slips, placements	Wizard inputs and selected covers, the full document (doc) with premium breakdown and customer data, risk details and requested covers
policies	<code>details</code> (risk and party details, renewal link), <code>doc</code> (policy document data carried from the quotation or placement)
insurer_offers	<code>taxes</code> , <code>deductibles</code> , <code>terms</code> of an insurer's answer
claims	<code>details</code> , <code>driver</code> , <code>third_party</code> , <code>policy_info</code> , <code>adjuster</code> , <code>settlement</code>
endorsements	<code>changes</code> (field changes and re-pricing), <code>endorsement_type_ids</code> , <code>completion</code>
renewals, renewal_quotes	Coverage details, accessories, order summary, limits, premium breakdown, taxes and rating
product_templates, product_components	<code>config</code> , <code>features</code> , <code>insurers</code> , <code>tags</code> ; component <code>data</code> (covers, rating tables, rules, the motor CTPL tariff)
master_records, master_types	<code>data</code> of generic masters (GIN index <code>master_records_data_idx</code> , <code>jsonb_path_ops</code>); <code>fields</code> and <code>unique_keys</code> of each master screen
app_settings	<code>value</code> : every configuration value
audit_log	<code>before_data</code> , <code>after_data</code> of every audited change
period end and bank reconciliation	<code>period_close_runs.steps</code> , <code>period_close_run_checks.detail</code> , <code>recurring_journals.lines</code> , <code>year_end_runs.checks</code> , <code>bir_2307_certificates.lines</code> , <code>bank_statement_formats.columns</code> , <code>bank_match_rules.params</code> , <code>bank_reconciliations.snapshot</code>
finance and reinsurance	<code>remittances.data</code> , <code>remittance_items.data</code> , <code>remittance_approvals.history</code> , <code>insurer_refund_credits.split</code> , <code>reinsurance_treaties.terms</code> , <code>incentive_programs.structure</code>
operations	<code>scheduled_jobs.params</code> , <code>job_runs.output</code> , <code>job_queue.payload</code> , <code>generated_reports.params</code> , <code>report_definitions.parameters</code>

3.4 Document numbering

Every document number comes from a series in the Document Numbering master (`document_numbering`, 52 series, Master > Document Numbering). `nextDocumentNumber(code, { db })` in `lib/numbering.js` calls the SQL function `next_document_number(code, branch, lob, date)`. The function locks the counter row of the series and period in `sequences` until the caller's transaction ends, so two users never get the same number and a rolled-back save gives its number back. `format_document_number()` applies the pattern: tokens `{PREFIX}` `{YYYY}` `{YY}` `{MM}` `{FY}` `{BRANCH}` `{LOB}` `{SEQ}`; all seeded series use `{PREFIX}-{YYYY}-{SEQ}`. The reset rule of a series is `yearly`, `fiscal_yearly`, `monthly` or `never`; the counter period is `2026`, `FY2027`, `2026-09` or `ALL`. The year is taken from the business date in `general.timezone`, so a document created at 00:30 Manila time on 1 January gets the new year. The next number of a series can only be moved forward (for example to continue from the legacy system at go-live). A trigger copies each prefix into the read-only setting `numbering.<code>.prefix`.

Counter (sequences.name)	Year	Last value (test data)	Counter	Year	Last value (test data)
adjustment	2026	4	receipt_txn	2026	23
bank_statement	2026	1	remittance	2026	17
bank_txn	2026	5	remittance_batch	2026	2
bordereau	2026	2	remittance_bill	2026	3

Counter (sequences.name)	Year	Last value (test data)	Counter	Year	Last value (test data)
cession	2026	13	remittance_exception	2026	4
disbursement_txn	2026	16	remittance_notice	2026	3
incentive_calc	2026	4	renewal	2026	1
invoice_list	2026	5	ri_reconciliation	2026	2
journal	2026	86	ri_recovery	2026	6
petty_cash	2026	3	settlement	2026	1
petty_cash_request	2026	2	transfer	2026	4
receipt	2026	18	voucher	2026	16

3.5 Audit, history and soft delete

Convention	Detail
Row stamps	<code>created_at</code> (107 tables), <code>updated_at</code> (83), <code>created_by</code> (84), <code>updated_by</code> (53). <code>updated_at</code> is set by the application; triggers maintain it on <code>users</code> , <code>roles</code> , <code>commission_rates</code> and <code>document_numbering</code> (<code>set_updated_at()</code>).
Audit log	<code>audit_log(id, at, user_id, username, entity, entity_id, action, before_data, after_data, ip)</code> , written by <code>lib/audit.js</code> from every create, update, approval, posting, run and sign-in related event; indexed on (<code>entity</code> , <code>entity_id</code>), (<code>at</code>) and (<code>username</code>).
Domain histories	<code>claim_history</code> , <code>claim_field_changes</code> , <code>renewal_activities</code> , <code>renewal_notices</code> , <code>collection_actions</code> , <code>remittance_approvals.history</code> , <code>period_status_history</code> , <code>bank_reconciliation_history</code> , <code>login_history</code> , <code>password_history</code> , <code>job_runs</code> .
Soft delete	<code>leads</code> , <code>quotes</code> and <code>bank_statements</code> carry <code>deleted_at</code> ; masters and product components use a <code>status</code> of <code>deleted</code> (unique indexes are partial: <code>WHERE status <> 'deleted'</code>). Financial documents are never deleted but cancelled or reversed (<code>receipts.cancelled_at</code> , journal reversal and correction vouchers, <code>bir_2307_certificates.cancelled_at</code>). A draft quotation is the only business record deleted physically.

3.6 Ledger integrity

Control	Detail
Posting rules	Every system journal is built by <code>postEvent()</code> from the posting rule of its business event (31 events, for example <code>policy.issue.broker_billed</code> , <code>receipt.apply</code> , <code>remittance.settlement</code> , <code>claim.funds_received</code> , <code>commission.clawback</code>). A rule has versions with an effective date (<code>posting_rules</code>) and lines (<code>posting_rule_lines</code>) that name either a GL account or an account role. Account roles resolve to GL accounts through the settings <code>accounting.account.<role></code> (Account Determination). A line marked per participant is split across the co-insurers by share, with the rounding remainder on the lead.
Balance and period trigger	Trigger <code>jv_check_posting</code> refuses to post a journal voucher with fewer than two lines, unbalanced, with a zero total, or in a period that does not accept postings.
Period status	<code>accounting_periods</code> holds periods 1 to 12 and the adjustment period 13 of each fiscal year with the status <code>open</code> , <code>soft_closed</code> , <code>closed</code> or <code>locked</code> . Trigger <code>accounting_period_lock_guard</code> refuses to reopen a period locked by the year-end close. Every change is written to <code>period_status_history</code> .
Opening balances	The year-end close writes the balance-sheet balances into <code>opening_balances</code> of the next year; the go-live import writes the old system's trial balance there (source <code>go-live:<date></code>). Reports read them through <code>pe_balance_before()</code> ; no opening journal is posted.

Control	Detail
Co-insurance	<code>risk_participants</code> holds the participants of a broker slip, quotation, placement or policy (one lead, shares total 100%, index <code>risk_participants_one_lead</code>); <code>receivable_participants</code> and <code>remittance_allocations</code> split bills and collections per insurer.
Account defaults	Trigger <code>gl_accounts_defaults</code> derives the normal balance of an account from its type.

3.7 Constraints and indexes

The schema has 448 indexes: 138 primary keys, 91 further unique indexes (business numbers, natural keys and partial uniqueness such as one open renewal per policy, one pending approval per entity, one active close run per period, one lead co-insurer per record), 33 partial indexes and one GIN index. Case-insensitive uniqueness uses expression indexes on `lower(...)`.

Migration `0146_foreign_key_indexes.sql` indexed the foreign keys of every table that grows (transactions, documents, lines and logs) and added the operational log indexes `job_runs (job_id, started_at)`, `email_outbox (status, created_at)`, `audit_log (at)` and `audit_log (username)`. The list was produced by `backend/scripts/fk-index-report.js`. 36 of the 234 foreign keys have no index whose first column is the foreign-key column; they are on small master and set-up tables (vehicle models, states, policy types, product components) or are maker-checker stamp columns (`approved_by`, `rejected_by`), left out on purpose. They are listed in document 03.

4 Entity-relationship diagrams

The diagrams are generated from the database catalogue. Each box lists the primary key (PK), up to two unique keys (UK), the foreign keys (FK; a red asterisk marks an FK without a supporting index) and the JSONB columns ({ }). Dashed boxes are tables of another domain. References to [users](#) (creator, owner, approver) are listed as columns but drawn only in the security diagram.

4.1 Sales and policy

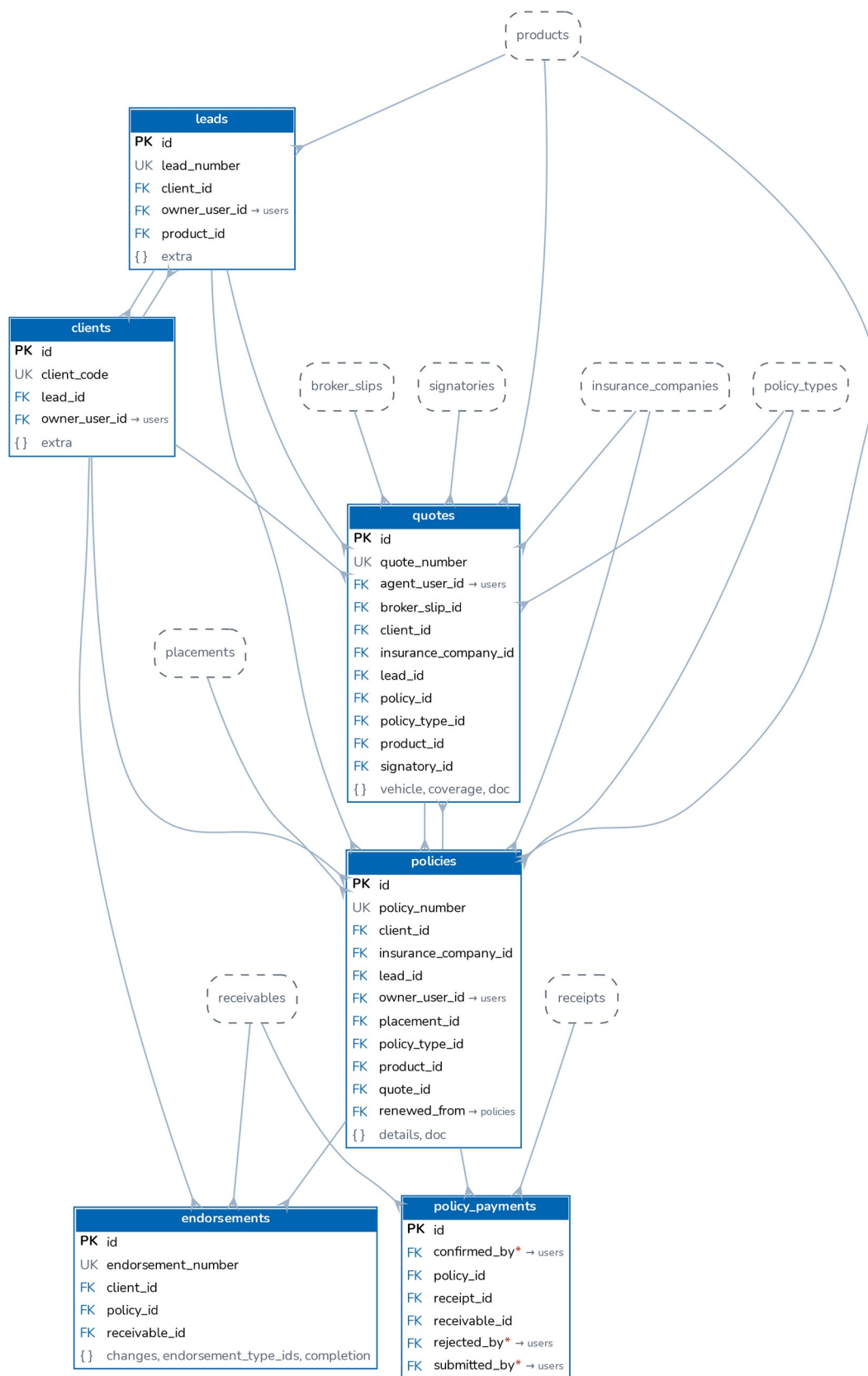


Figure 1. Sales and policy: leads, clients, quotations, policies, endorsements, payment captures

A lead becomes a client when a quotation is converted (`clients.lead_id` is unique when set). A quotation references the lead or client, product, policy type, insurer and signatory; conversion creates the policy

(policies.quote_id, quotes.policy_id). A renewed policy points to its predecessor (policies.renewed_from, renewed_to). Endorsements belong to a policy and may raise their own bill (endorsements.receivable_id). policy_payments holds payments captured by front-office staff until Accounting confirms them and issues a receipt.

4.2 Placement and co-insurance

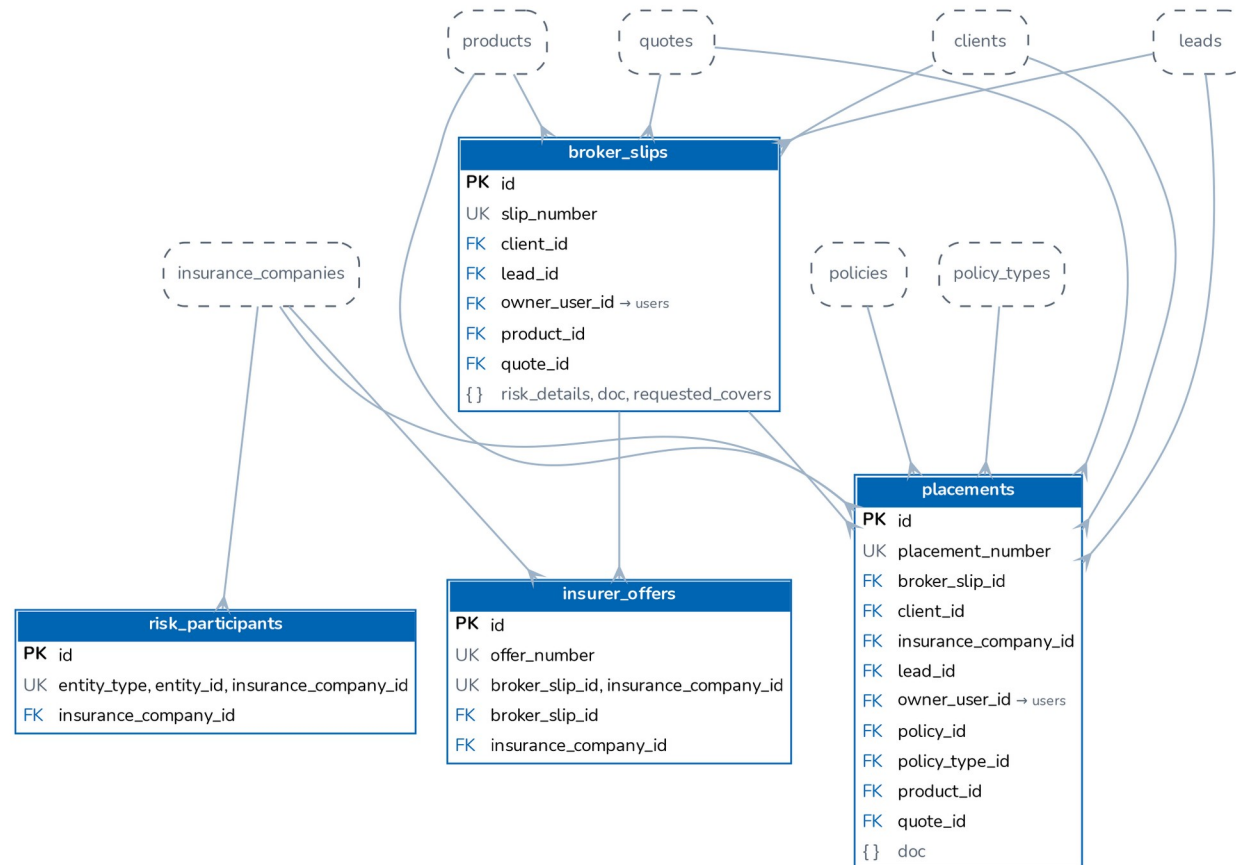


Figure 2. Placement: broker slips, insurer offers, placement slips and co-insurance participants

A broker slip is created from a lead or client and sent to several insurers; each answer is an **insurer_offers** row (premium, rate, taxes, deductibles, validity, share offered) or a decline. The chosen offers become a quotation or a placement slip (**placements.broker_slip_id**, **placements.quote_id**). A placement is sent to the lead insurer and the co-insurers; each participant binds with its own reference and the policy is issued when all have bound (**placements.policy_id**). **risk_participants** is shared by all four record types through **entity_type** and **entity_id**.

4.3 Renewals

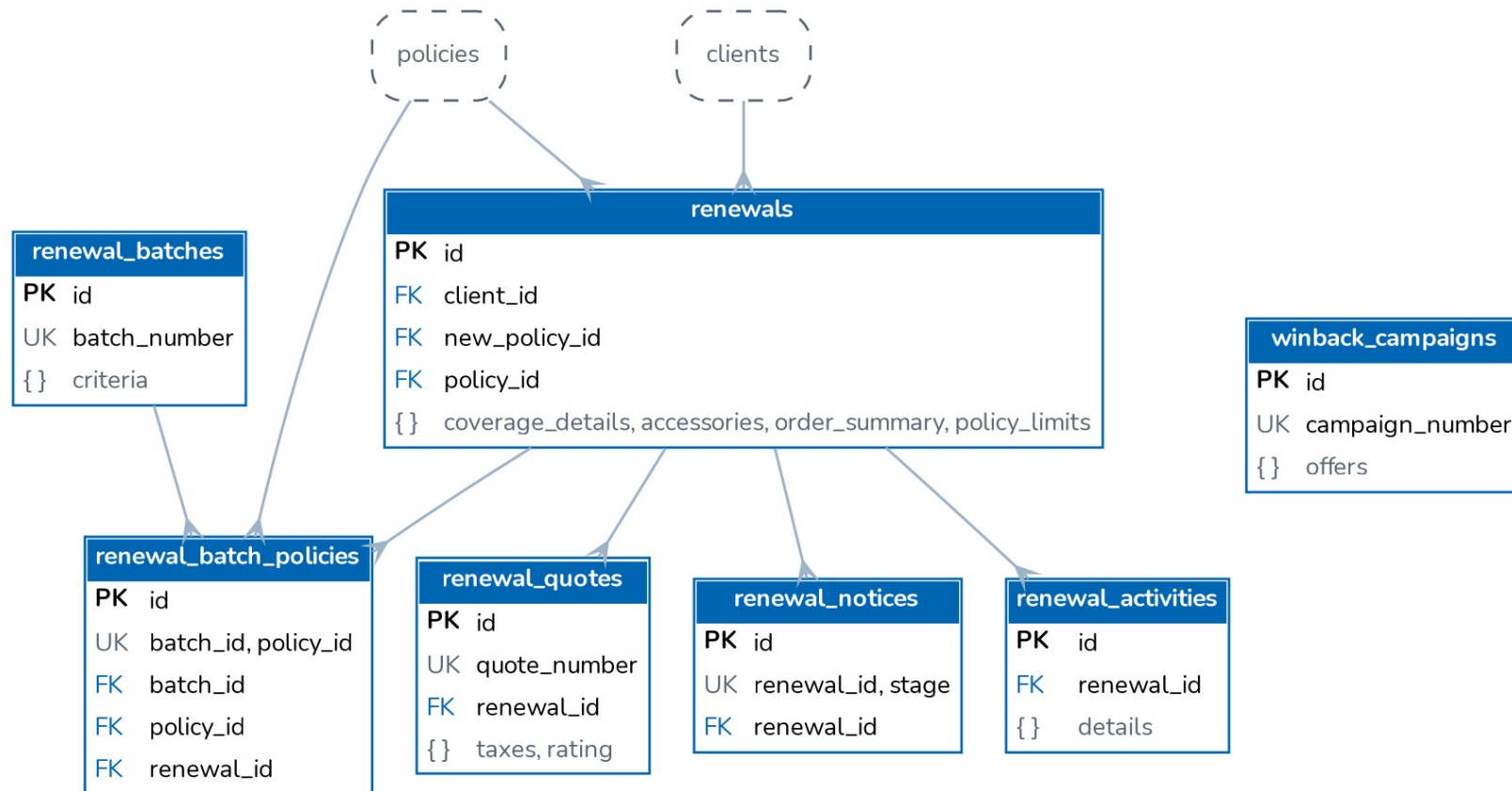


Figure 3. Renewals: renewal cases, quotes, notices, activities, batches, win-back campaigns

4.4 Billing, receipts and collections

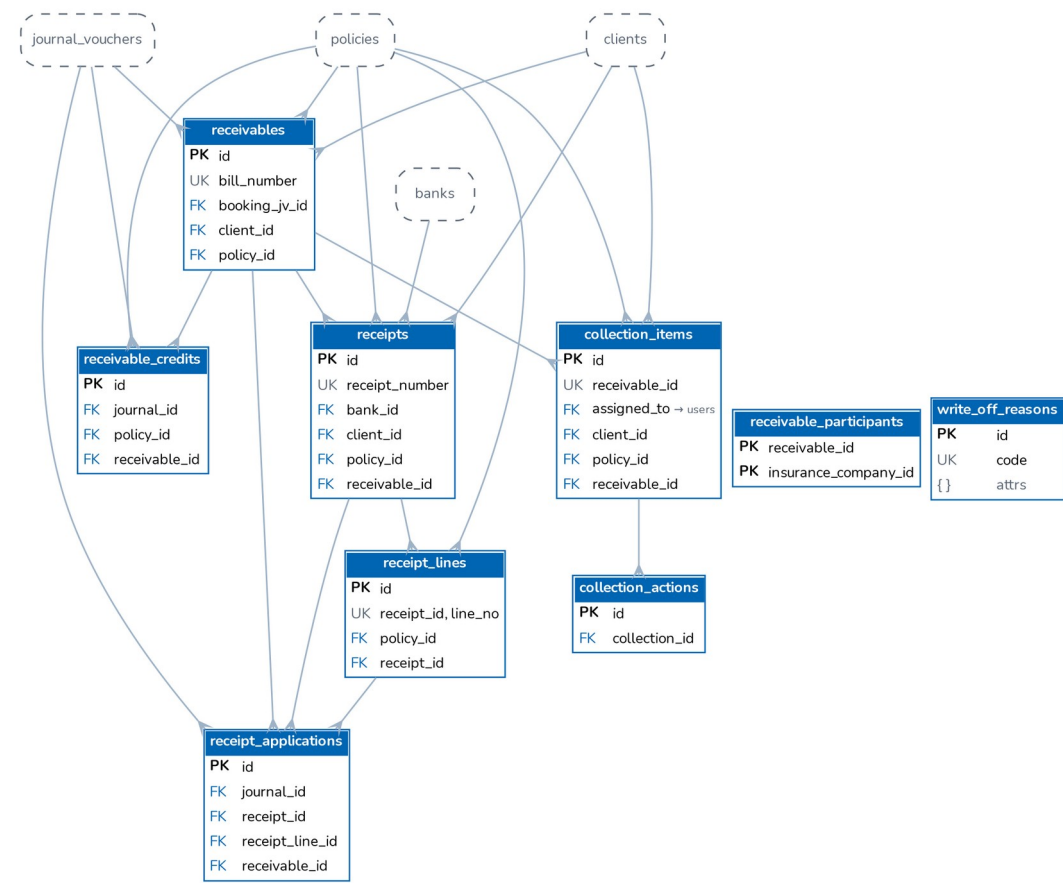


Figure 4. Billing, receipts, credits, co-insurance split and collections

A policy, endorsement or renewal raises a receivable (bill) with its booking journal (*receivables.booking_jv_id*). A receipt has lines per policy, and receipt applications allocate the lines to bills and link the posting journal. Return premium and cancellations reduce a bill through *receivable_credits*. Collection items track the follow-up of open bills; write-offs use the reasons and GL accounts of *write_off_reasons*.

4.5 General ledger and posting rules

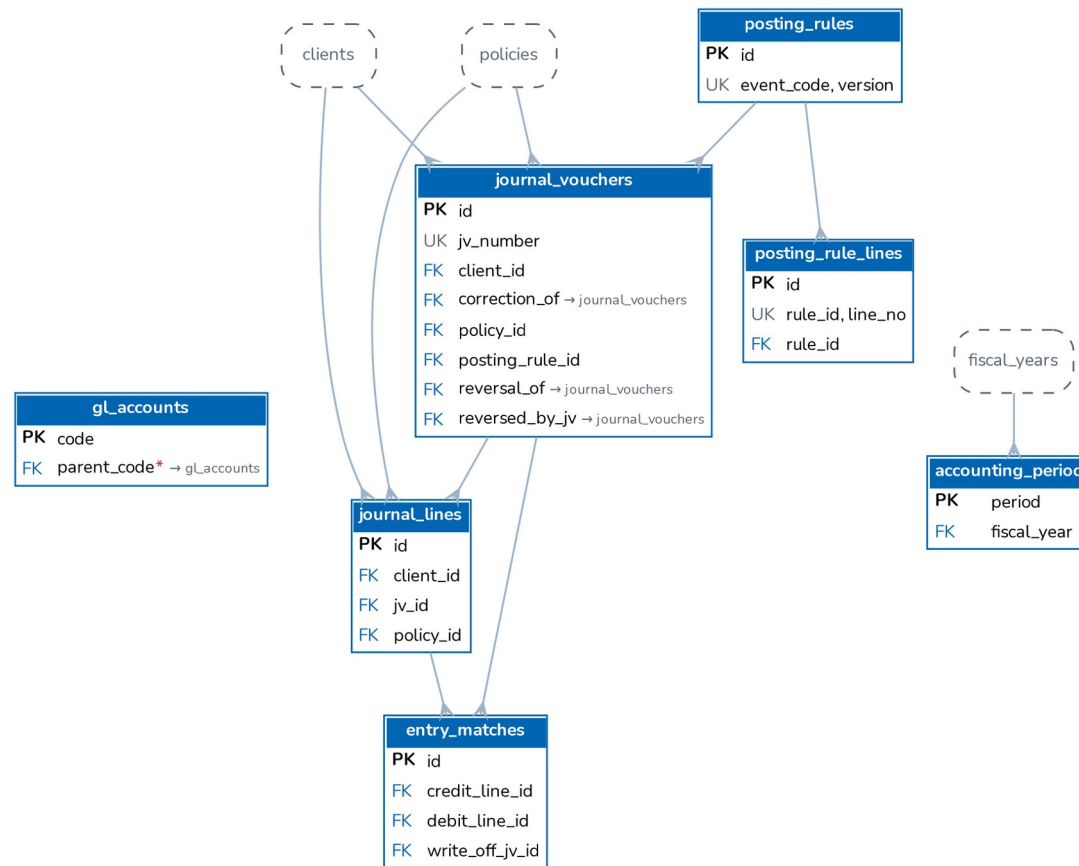


Figure 5. General ledger: chart of accounts, journals, periods, matching and posting rules

Journal vouchers hold balanced journal lines coded to the chart of accounts (**gl_accounts**). Open-item accounts are reconciled by **entry_matches** (debit line to credit line). Journals point back to their source document through **journal_vouchers.reference_type** and **reference_id**.

4.6 Period end and tax

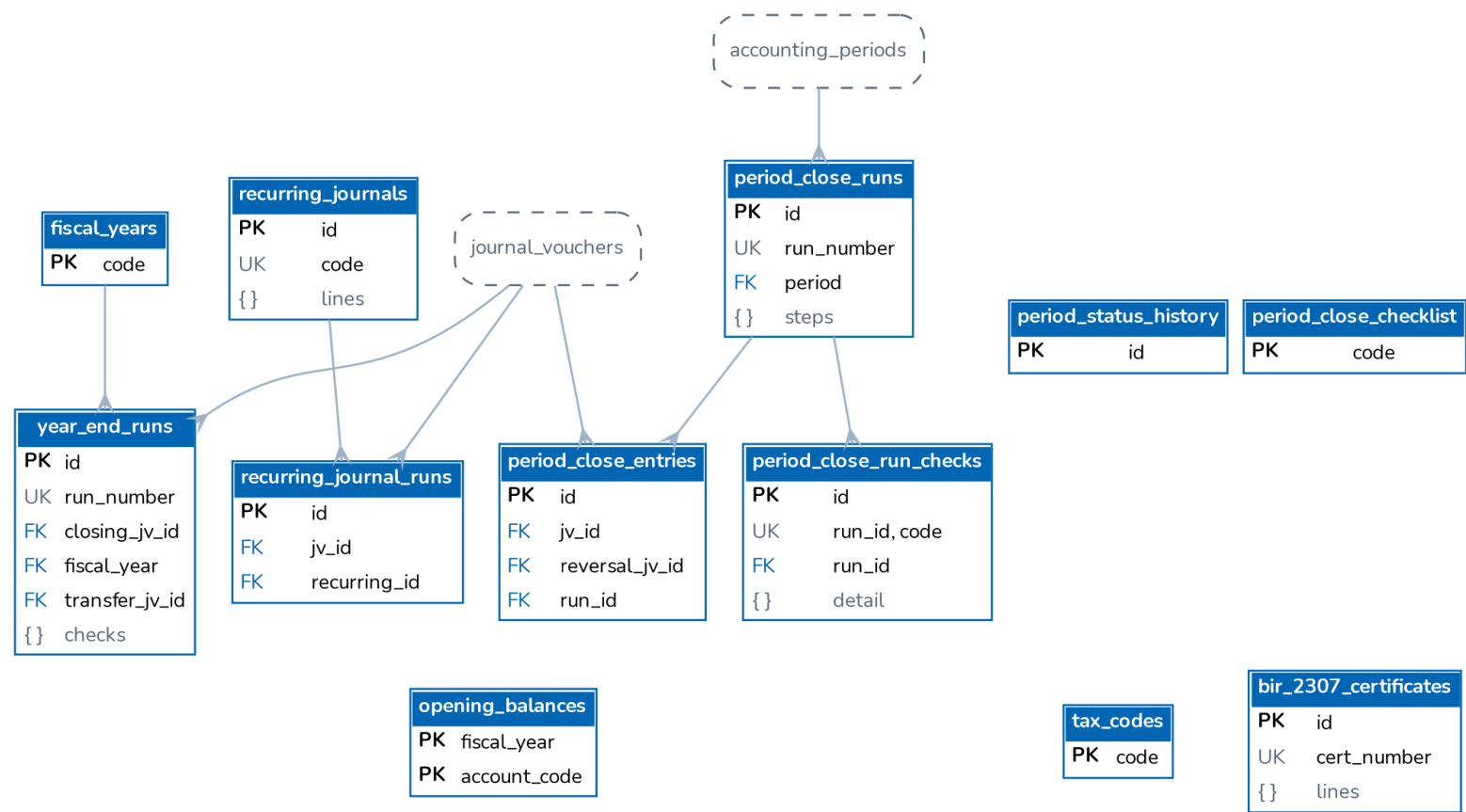


Figure 6. Period end: fiscal years, close runs, recurring journals, year-end, tax codes and BIR 2307

4.7 Bank reconciliation

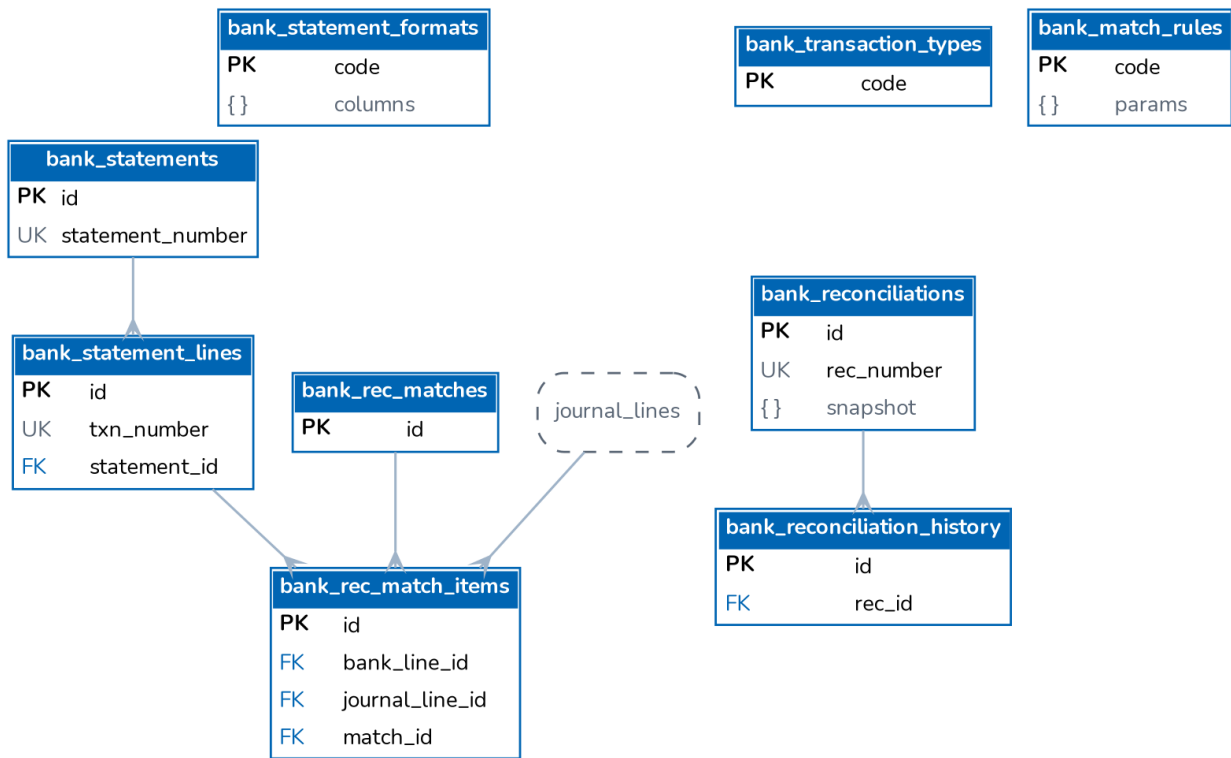


Figure 7. Bank reconciliation: statement formats, statements and lines, matches, reconciliations

A bank statement is imported with a statement format into `bank_statements` and `bank_statement_lines` (a line hash refuses duplicates). Matches link bank lines to cash-account journal lines (`bank_rec_match_items` with a *side*). A monthly reconciliation (`bank_reconciliations`) stores the statement figures and, once approved, locks the matches cleared up to the period end. The bank account itself is a master record linked to a GL cash account.

4.8 Disbursements and petty cash

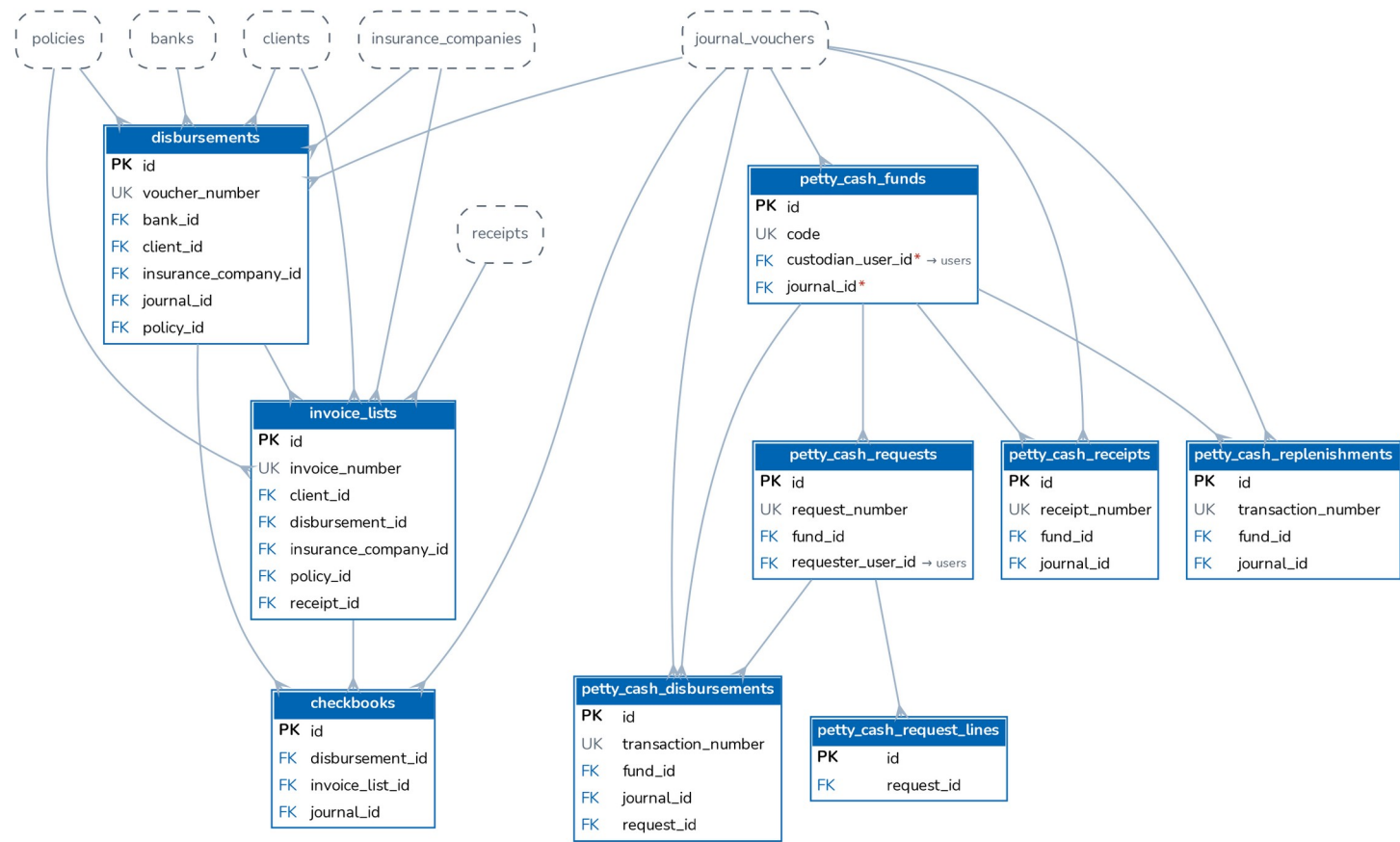


Figure 8. Disbursements (payment vouchers, invoice lists, cheques) and petty cash

4.9 Commission and incentive

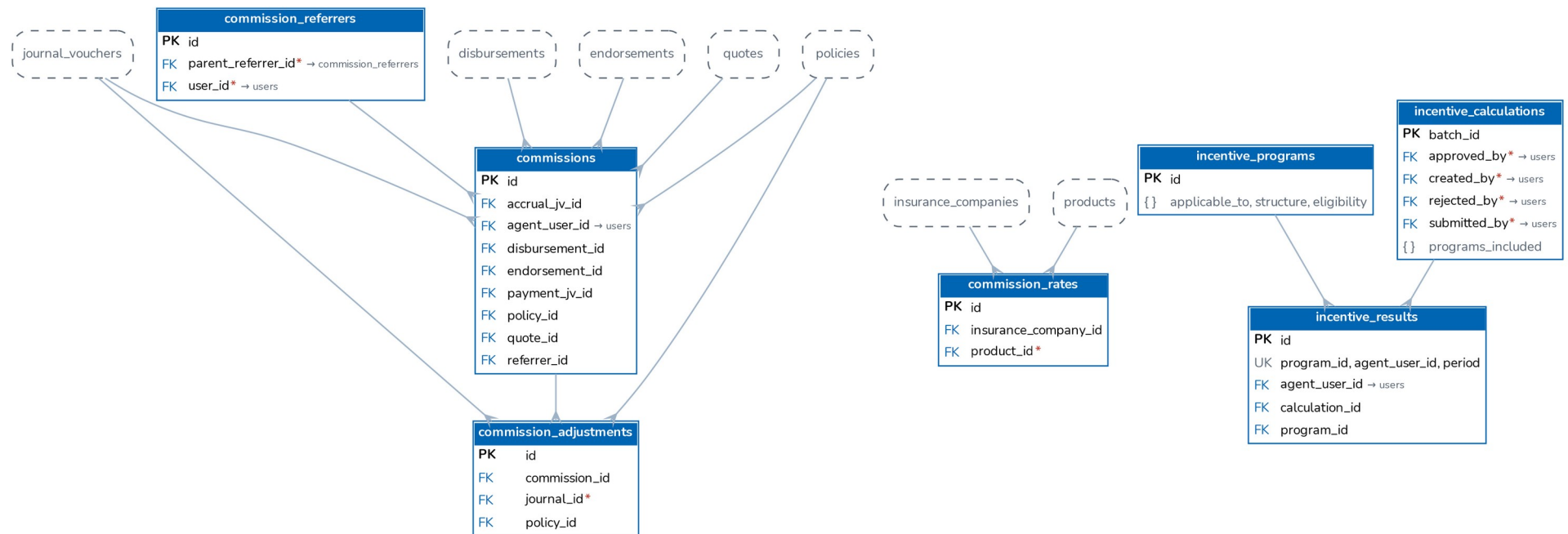


Figure 9. Commission lines, adjustments, the rate matrix, referrers and incentive programmes

Commission lines are created per policy, endorsement and referrer in the referral chain (*commissions_policy_referrer_uq*), link to the accrual and payment journals and to the payment voucher that paid them. *commission_adjustments* records the reversal or clawback when premium is returned. The rate of a new policy comes from *commission_rates* (insurer, product, line, policy type, effective dates), else the insurer's default rate. Referrers form a hierarchy (*parent_referrer_id*); they do not sign in.

4.10 Claims

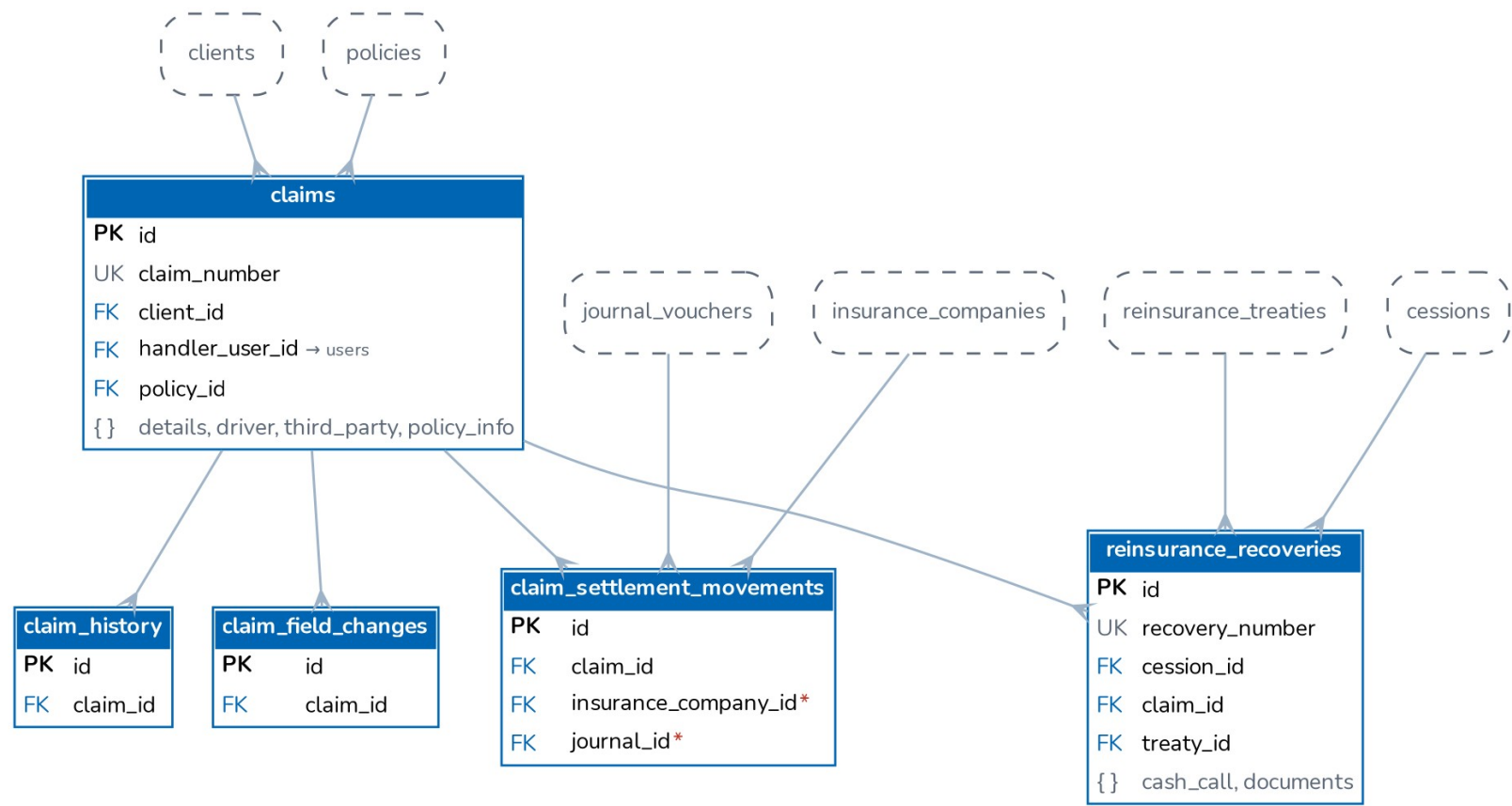


Figure 10. Claims, claim history, field-level changes, settlement cash and reinsurance recoveries

4.11 Remittance and direct bill

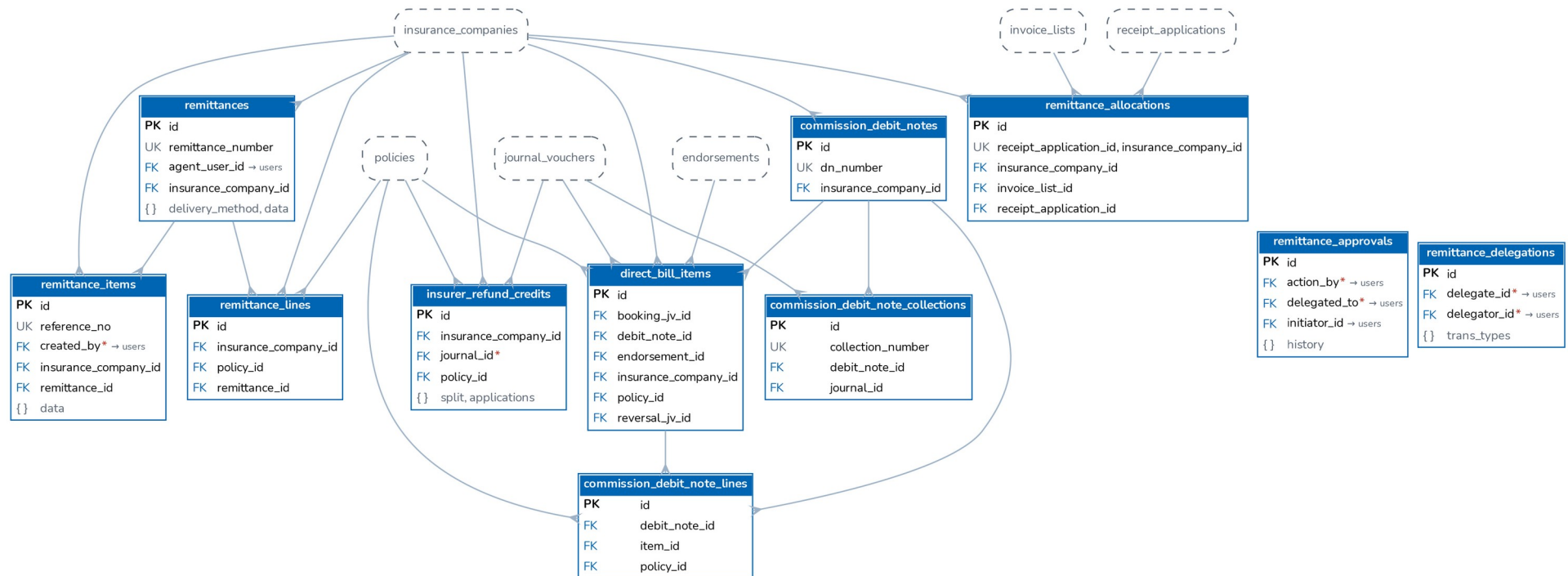


Figure 11. Remittances to insurers, co-insurance allocations, insurer refunds and direct-bill debit notes

Broker-billed premium is remitted to insurers through **remittances** with **remittance_lines** per policy; settlements, adjustments and exceptions are **remittance_items**; **remittance_approvals** is a multi-level approval register (entity, levels, SLA, history). For a co-insured policy **remittance_allocations** splits each collection by insurer share. **insurer_refund_credits** holds refunds due from an insurer on returned premium already remitted; they are netted against the next remittance. Direct-billed policies produce **direct_bill_items** (commission receivable) billed to the insurer on **commission_debit_notes** and settled by **commission_debit_note_collections**.

4.12 Reinsurance

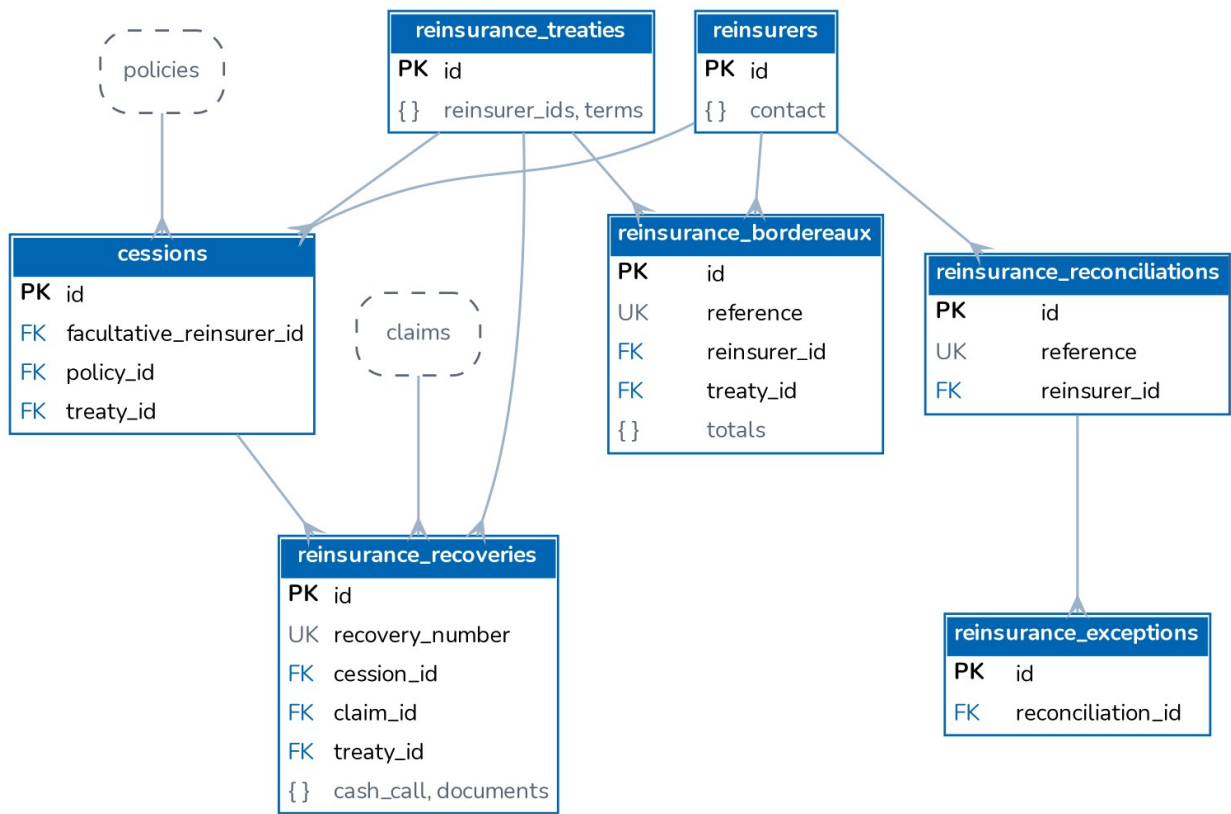


Figure 12. Reinsurers, treaties, cessions, recoveries, bordereaux and reconciliations

4.13 Configuration, security and platform

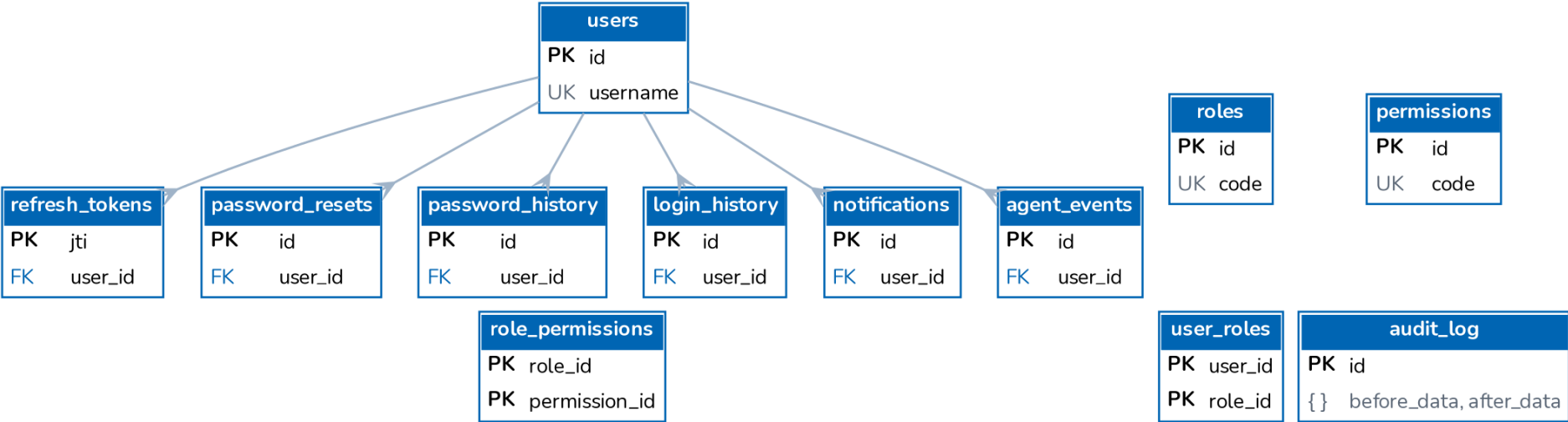


Figure 13. Security and identity tables

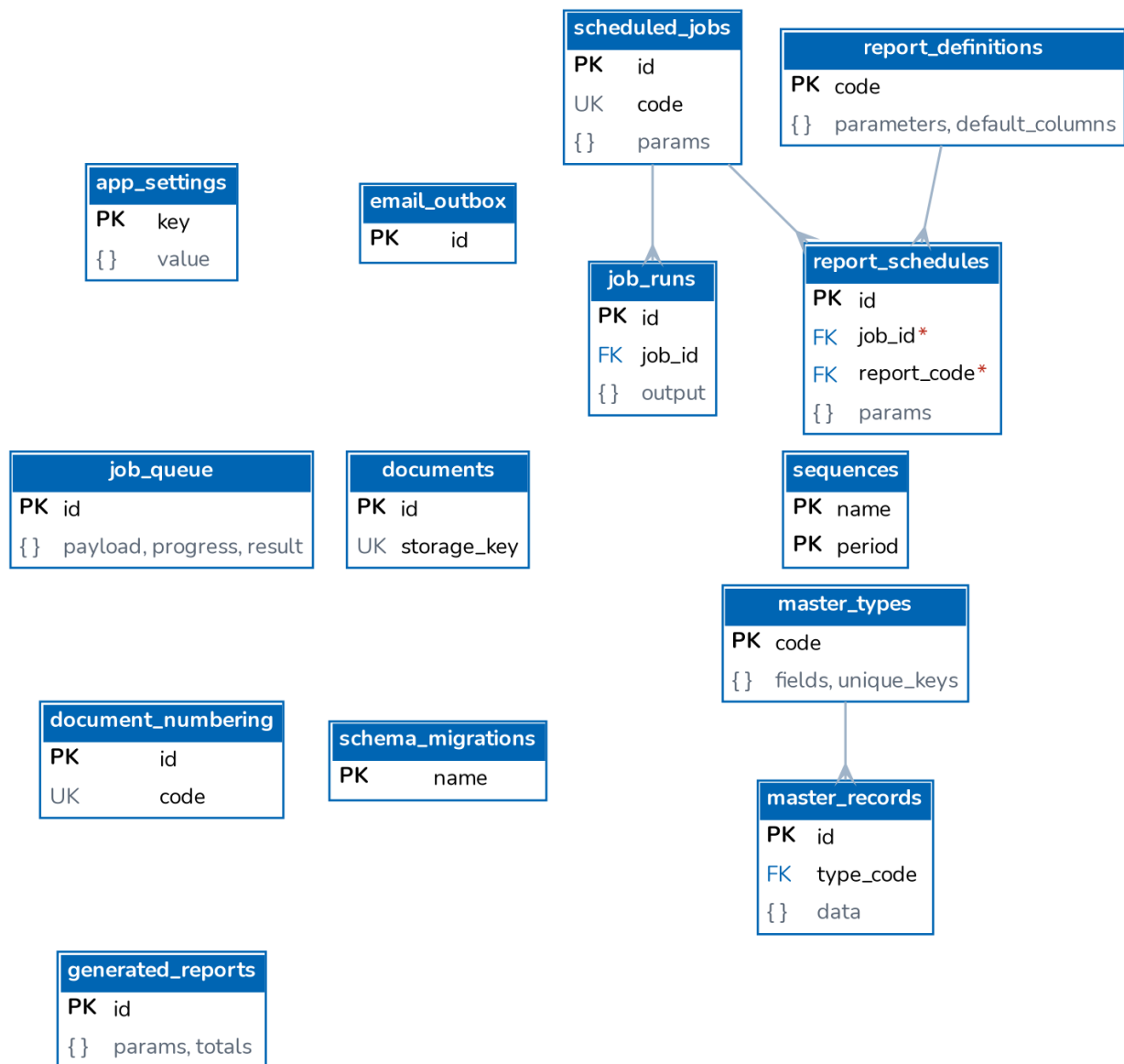


Figure 14. Configuration and platform services tables

4.14 Reference masters and product configurator

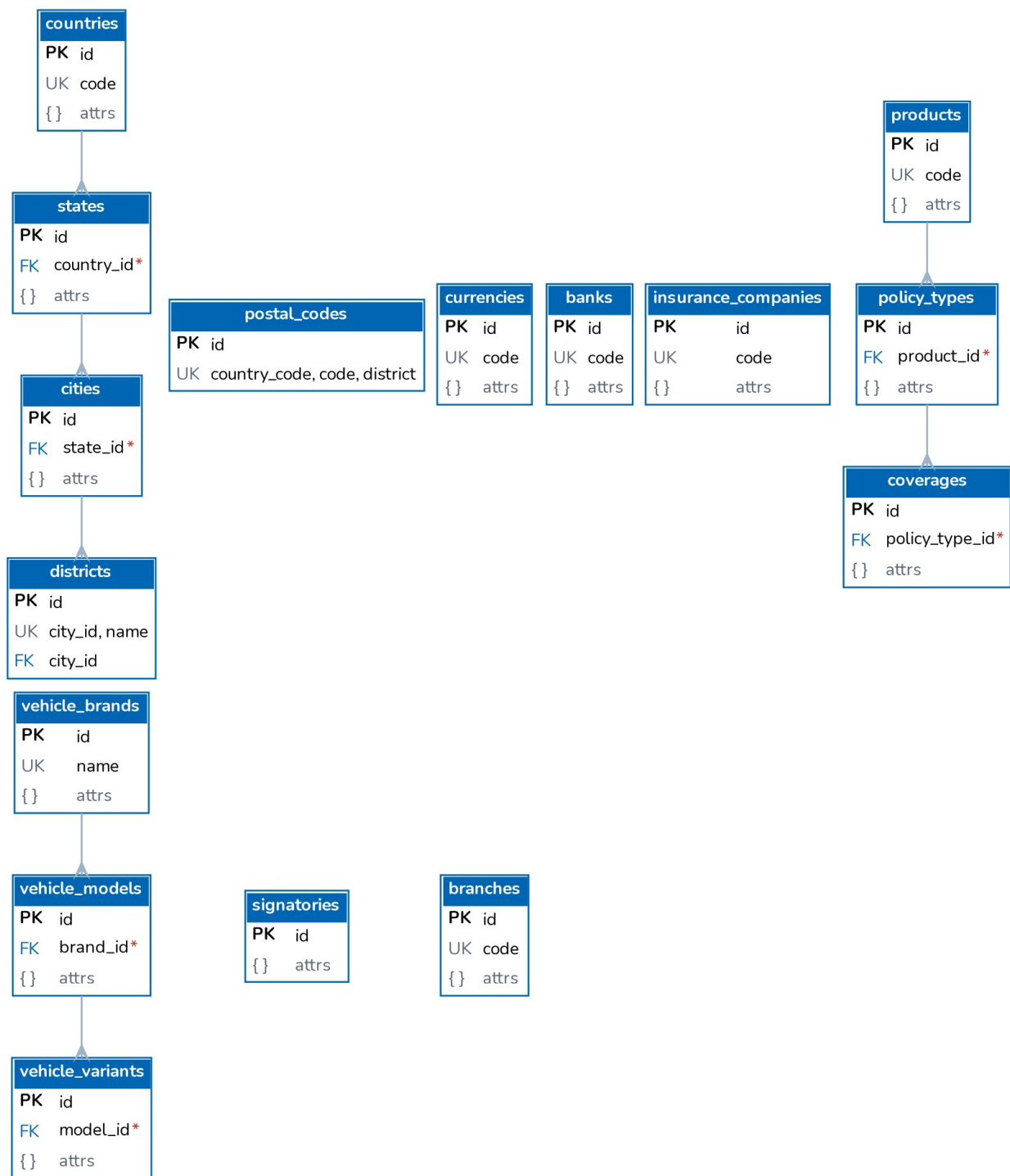


Figure 15. Reference masters

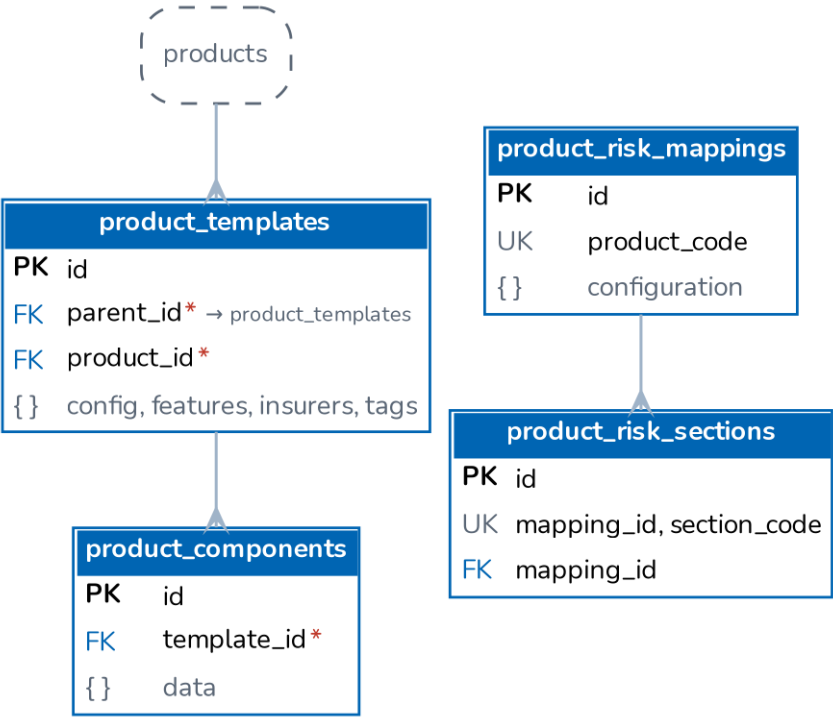


Figure 16. Product configurator

5 Migrations

Aspect	Implementation (<code>backend/src/db/migrate.js</code>)
Format	Plain SQL files <code>NNNN_name.sql</code> in <code>backend/src/db/migrations</code> , applied in file-name order: 0001-0004 core, masters, operations, finance; 0010-0021 sales, claims, renewals; 0030-0034 ledger, receipts, disbursements, commission, petty cash; 0040-0050 configuration, product configurator, remittance, reinsurance, incentive, reports; 0060-0097 security and later changes; 0110-0113 branding and the Company letterhead; 0115-0117 role inheritance and bank reconciliation; 0120-0127 co-insurance, document numbering, commission rates, insurer credit terms, placement; 0130-0136 posting rules, period end, tax codes; 0140-0152 broker roles, business-date defaults, foreign-key indexes, accounting follow-ups, go-live imports and the e-mail sender.
Tracking	Table <code>schema_migrations(name, applied_at)</code> ; a file is applied once.
Execution	On every API start (<code>server.js</code>), before the seed. The migrate step first takes the advisory lock <code>hashtext('brokerverse.migrate')</code> , <code>hashtext('schema_migrations')</code> ; instances that start together wait (up to <code>MIGRATION_LOCK_TIMEOUT_MS</code> , 10 minutes) and then find nothing to do. Each pending file runs in its own transaction with its <code>schema_migrations</code> row; on error the transaction is rolled back and the API does not start. <code>npm run migrate</code> runs the same code; <code>npm run db:reset</code> drops and recreates the schema (development only).
Idempotency	Files use <code>CREATE TABLE IF NOT EXISTS</code> , <code>ADD COLUMN IF NOT EXISTS</code> , <code>CREATE INDEX IF NOT EXISTS</code> , <code>ON CONFLICT DO NOTHING</code> and guarded updates, so they are safe on a database that already holds data. An applied file is never edited; fixes go forward in a new file.
Readiness	<code>GET /api/health</code> answers 503 until no migration is pending, so a load balancer does not route to an instance with an older schema.
Direction	Forward only. Migrations only add; rolling back data means restoring a snapshot (document 09).

Recommended / to be confirmed by the business and DevOps: Keep destructive changes (column drops, type changes) in a separate release after the code that no longer uses them is live. Start one instance first when a release contains migrations (`deploy/REFERENCE.md`, Upgrading).

6 Seed data

`backend/src/db/seed.js` runs on every start after the migrations and with `npm run seed`. Every step is idempotent: rows are inserted by natural key or fixed id, so administrator edits are kept.

1. In code: the roles, permissions and grants, and the administrator **BrokerVerse** (password from `ADMIN_PASSWORD`; without it a random password is generated, logged once and must be changed at the first sign-in).
2. `seeds/settings.json` (configuration keys, inserted only if missing) and `seeds/jobs.json` (the scheduled jobs; the period-end and bank-matching jobs come from migrations).
3. Encryption of any two-factor secret still stored in clear (upgrade path).
4. The SQL files `seeds/*.sql` (**reference data**, always) and `seeds/sample/*.sql` (**sample data**, only when `SEED_SAMPLE_DATA` is on), interleaved by file name so a sample file runs after the reference file it depends on.

Class	Content
Reference	Settings (351 keys); 7 roles and 56 permissions; Philippine geography and postal codes; currencies, banks, insurers, products and policy types; vehicle makes, models and variants; covers; head office branch; the Company master with the letterhead company; master screen registry and reference master records; chart of accounts (129 accounts in this database) with main and sub account masters; posting rules for 31 events and Account Determination; 52 document number series; tax codes (VAT, expanded withholding with ATC, DST, LGT); the month-end checklist; bank statement formats (three Philippine bank layouts and a generic one); product templates with the motor and CTPL tariff; report catalogue (39 reports); direct-bill and security configuration.
Sample	Fictional leads, clients, quotations, broker slips and placements, policies, endorsements, claims, renewals, bills, receipts, collections, disbursements, petty cash, commissions and referrers, remittances, reinsurance, incentives, journal vouchers, bank statements, demo branches, signatories and master records, and six sample users (five Sales & Marketing users and one Accounting approver).

`SEED_SAMPLE_DATA` accepts true or false (1 or 0, yes or no, on or off); unset, it is on in development and test and off with `NODE_ENV=production`. A database seeded with sample data is cleaned with `npm run purge:sample` (document 10).

7 Time zone, date and amount handling

Topic	Design
Storage	Instants in <code>timestamp_tz</code> (stored in UTC); calendar dates in <code>date</code> . The database time zone is Asia/Manila; the containers run in UTC.
Business date	<code>lib/dates.js</code> : "today" is the calendar date in <code>general.timezone</code> (seeded Asia/Manila), not the server's UTC date. <code>businessDate()</code> , <code>calendarPeriod()</code> , the report engine, document numbering and the column defaults (<code>numbering_business_date()</code> , migration 0145) use the configured zone, so between 00:00 and 08:00 Manila time the system never uses yesterday's date.
Job times	The scheduler reads cron expressions in <code>general.timezone: 0 6 * * *</code> runs at 06:00 Manila time.
Display	The front end formats dates with <code>general.date_format</code> (DD/MM/YYYY) and amounts with the configured currency and locale (en-PH grouping); the server uses <code>formatMoney()</code> and the PDF engine the same settings.
Money	<code>numeric</code> in the database, rounding by <code>round2()</code> ; taxes (VAT 12%, DST 12.5%, LGT 0.75%, expanded withholding by ATC) and commission rates come from settings, the tax codes and the rate matrix.

Note: Amounts are converted from `numeric` to JavaScript double-precision numbers. This is exact for amounts of up to about 90 trillion with two decimals, far above broking volumes, provided every arithmetic result is rounded with `round2()`, which the ledger and document code does.

8 Related documents

This document is one of eleven that together form the BrokerVerse OOTB Solution Architecture. They are maintained and issued together (source and build: [docs/architecture/README.md](#)).

No.	Document	File
01	Solution Component Diagram and Application Architecture	01_BrokerVerse_Solution_Component_Diagram_Application_Architecture.pdf
02	Database Design (this document)	02_BrokerVerse_Database_Design.pdf
03	Database Inventory	03_BrokerVerse_Database_Inventory.pdf
04	Technology Stack	04_BrokerVerse_Technology_Stack.pdf
05	Shared Service Components	05_BrokerVerse_Shared_Service_Components.pdf
06	Capacity and Performance	06_BrokerVerse_Capacity_and_Performance.pdf
07	High Availability and Resiliency	07_BrokerVerse_High_Availability_and_Resiliency.pdf
08	RTO and RPO	08_BrokerVerse_RTO_and_RPO.pdf
09	Backup and Recovery	09_BrokerVerse_Backup_and_Recovery.pdf
10	Data Archival, Housekeeping and Restoration	10_BrokerVerse_Data_Archival_Housekeeping_and_Restoration.pdf
11	Monitoring	11_BrokerVerse_Monitoring.pdf

9 Glossary

Terms and abbreviations used in this document.

Term	Meaning
Account determination	Settings accounting.account.<role> that give the GL account of each account role (cash in bank, premium receivable, due to insurer, commission income ...) used by the posting rules.
Advisory lock	PostgreSQL application-level lock (pg_try_advisory_lock, pg_advisory_lock) used so a job runs on one API instance at a time and only one instance applies migrations.
API	Application programming interface; here the BrokerVerse REST API under /api served by the Node.js backend.
ATC	Alphanumeric tax code of the BIR for a type of withholding, printed on BIR Form 2307.
Audit log / audit trail	Table audit_log: who changed which record, when, from which IP, with the before and after values.
BIR	Bureau of Internal Revenue (Philippines): tax authority whose rules drive record keeping and withholding tax certificates.
Bordereau	Periodic list of ceded premiums or claims sent to a reinsurer.
Broker slip	Request for quotation (BS-) that the broker sends to several insurers for one client risk; the answers are recorded as insurer offers.
Cession	The share of a policy's risk and premium passed to a reinsurer under a treaty or facultatively.
Co-insurance	Placement of one risk with several insurers, one of them the lead, each taking a share; premium, commission and claims are split by share.
Company master	Master record of the broker company; the primary company is the letterhead printed on every document and report.
Cron	Time-based schedule expression (minute hour day month weekday) used by the scheduled jobs.
CTPL	Compulsory Third Party Liability motor insurance (Philippines).
Debit note (DN)	Commission debit note raised by the broker to an insurer for direct-billed policies.
Direct bill	Billing mode in which the insurer collects the premium and the broker bills the insurer for its commission.
DST	Documentary Stamp Tax (Philippines), part of the gross premium.
Endorsement	Change to an issued policy (details, cover, premium or cancellation).
Fiscal year	Accounting year (FY2026) with twelve periods and an adjustment period 13 used by the year-end close.
FK	Foreign key: column that references the primary key of another table.
GIN	Generalised inverted index (PostgreSQL), used for JSONB searches.
Housekeeping	Daily job that deletes operational rows (job runs, sent e-mails, sign-in history, expired tokens, read notifications) after the retention days in System Settings.
JSONB	PostgreSQL binary JSON column type, used for flexible document parts of records.
LGT	Local Government Tax (Philippines), part of the gross premium.
Maker-checker	Control in which a record created or submitted by one user must be approved by a different user.
Migration	Numbered SQL file in backend/src/db/migrations applied once, in order, on API start.
OR	Official receipt issued for premium received.
PK	Primary key: column(s) that uniquely identify a row.
Placement slip	Firm order (PS-) sent to the lead insurer and co-insurers; each participant binds before the policy is issued.

Term	Meaning
Posting rule	Versioned definition of the journal lines for one business event (policy issued, receipt applied ...); every system journal is built from it.
RDS	Amazon Relational Database Service, the managed PostgreSQL target.
Receivable / bill	Premium bill (INV-) raised when a policy, endorsement or renewal is issued.
Remittance	Payment of collected premium, net of commission, from the broker to the insurer.
RPO	Recovery point objective: the maximum acceptable data loss, measured in time.
RTO	Recovery time objective: the maximum acceptable time to restore service.
Seed	Idempotent data load run on every API start: reference data always, sample data only when SEED_SAMPLE_DATA is on.
SLA	Service level agreement / target time, e.g. for remittance approvals.
VAT	Value-added tax (12 % in the Philippines).
Year-end close	Run (YEC-) that posts the closing entries in period 13, carries the balances into the next year and locks the year.
zod	TypeScript-first schema validation library used to validate request bodies and parameters.